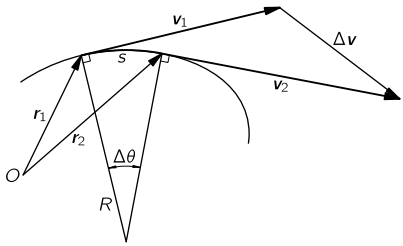

Figure 1-7 A curved trajectory.

$\mathbf{v}_1$; that is, we draw $\Delta\mathbf{v}$ as the difference of the two vectors, right? *No!* That only works when the *tails* of the vectors are in the same place! It has no meaning if we move the vector somewhere else and then draw a line across, so watch out! We have to draw a new diagram to subtract the vectors. In Figure 1-8, $\mathbf{v}_1$ and $\mathbf{v}_2$ are both drawn parallel and equal to their counterparts in Figure 1-7, and now we can discuss the acceleration. Of course the acceleration is simply $\Delta\mathbf{v}/\Delta t$. It is interesting to note that we can compose the velocity difference out of two parts; we can think of acceleration as having *two components*, $\Delta\mathbf{v}_{||}$ in the direction tangent to the path and $\Delta\mathbf{v}_{\perp}$ at right angles to the path, as indicated in Figure 1-8. The acceleration tangent to the path is, of course, just the change in the *length* of the vector, i.e., the change in the *speed v*:

$$a_{||} = dv/dt. \tag{1.15}$$

The other component of acceleration, at right angles to the curve, is easy to calculate, using Figures 1-7 and 1-8. In the short time Δt let the change in angle between $\mathbf{v}_1$ and $\mathbf{v}_2$ be the small angle $\Delta\theta$. If

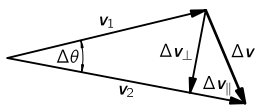

Figure 1-8 Diagram for calculating the acceleration.